

Loop Invariants and the Correctness Proof

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1 Loop Invariant ?

Loop invariant .

1. Initialization

loop invariant .

2. Maintenance

invariant , .

3. Termination

, invariant .

2 Insertion Sort Loop Invariant

Insertion Sort for ,
A[1 ... j-1] , .

2.1 Initialization

j = 2 , A[1 ... j-1] A[1] .
, invariant .

2.2 Maintenance

A[j]
(A[j-1], A[j-2], ...)
A[j] A[1 ... j] .
A[1 ... j-1] .

2.3 Termination

for $j = n+1$.
invariant $A[1 \dots n]$.
 , **Insertion Sort** .